

Programme	B. Sc. Computer Science				
Course Title	Client-Server Architecture				
Type of Course	Major				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Knowledge in Fundamentals of Network and Operating System				
Course Summary	The syllabus is prepared with the view of preparing the BSc Computer Science Graduates to build effective Client/Server applications. This course aims at providing a foundation in decentralized computer systems, using the client/server model. The course content is decided to cover the essential fundamentals which can be taught within the given slots in the curriculum.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basics of client/server systems and the driving force behind the development of client/server systems.	U	C	Instructor-created exams / Quiz
CO2	Outline the architecture and classifications of client/server systems	U	C	Instructor-created exams / Quiz
CO3	Choose the appropriate client/server network services for a typical application	U	P	Instructor-created exams / Quiz
CO4	Describe management services	U	C	Instructor-created exams / Case studies
CO5	Describe issues in network	U	P	Instructor-created exams / Quiz Case studies
CO6	Apply various services and support	U	P	Instructor-created exams / Quiz /Case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Client/Server computing		12	15
	1	Introduction to Client/Server computing - Driving forces behind Client/Server, Client/Server development tools	2	
	2	Development of client/server systems, Client/Server security	2	
	3	Organizational Expectations, Improving performance of client/server applications	2	
	4	Single system image, Downsizing and Rightsizing	3	
	5	Advantages of client server computing, Applications of Client/Server	3	
II	Client/Server Application Components		12	15
	6	Classification of Client/Server Systems- Two-Tier Computing, Middleware, Three-Tier Computing	2	
	7	Model View Controller (MVC)	1	
	8	Principles behind Client/Server Systems	3	
	9	Client/Server Topologies	3	
	10	Existing Client/Server Architecture.	1	
	11	Architecture for Business Information System	2	
III	Client/Server Systems Development		12	20
	12	Client- Services, Request for services, RPC, Windows services, Print services, Remote boot services, other remote services, Utility Services.	2	
	13	Dynamic Data Exchange (DDE).	2	
	14	Object Linking and Embedding (OLE).	2	
	15	Common Object Request Broker Architecture (CORBA).	2	
	16	Server- Detailed server functionality	2	
	17	Network operating system, Available platforms, Server operating system.	2	
IV	Client/Server Systems Development		12	20
	18	Services and Support- System administration, Availability, Reliability, Scalability, Observability, Agility, Serviceability.	2	
	19	Software Distribution, Performance, Network management.	2	
	20	Remote Systems Management- RDP, Telnet, SSH, Security.	3	
	21	LAN and Network Management issues, Training, Connectivity,	2	
	22	Communication interface technology, Inter process communication,	3	
V	Open Ended Module		12	
	CASE STUDY: Client Server Architecture Generic Client/Server Classes Client/Server Communication via Sockets The Server Protocol The Client Protocol A Two-Way Stream Connection			

1. Patrick Smith & Steave Guengerich, "Client / Server Computing", PHI
2. Dawna Travis Dewire, "Client/Server Computing", TMH
3. Jeffrey D.Schank, "Novell's Guide to Client-Server Application & Architecture" Novell Press
4. Robert Orfali, Dan Harkey, Jeri Edwards, Client/Server Survival Guide, Wiley-India Edition, Third Edition

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO1	-	2	-	-	-	-						
CO2	-	3	-	-	-	-						
CO3	-	2	1	-	-	-						
CO4	-	2	1	1	1	1						
CO5	-	2	1	-	-	-						
CO6	-	2	1	-	1	1						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	